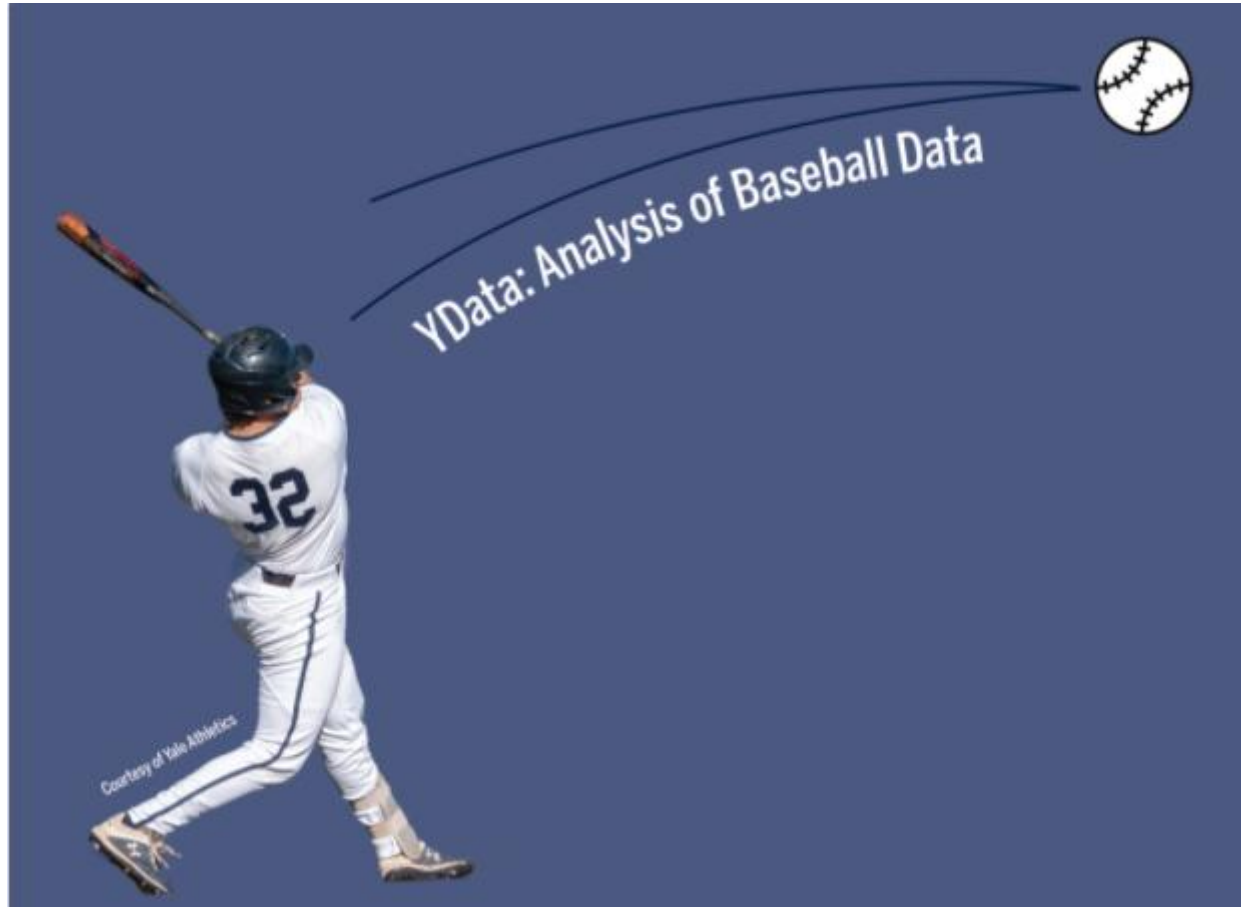


S&DS 1730



Class 7: Probability models and expected values

Overview

Discussion of chapter 6 of Astrobball

Probability distributions

Brief discussion of hypothesis tests

Expected value and calculating the run expectancy matrix

Sam Larson discussing his work for the Boston Red Sox

Final projects!

Use the Data Science methods you have learned in this class to answer a question related to baseball!

- Ex 1: Is it true that some baseball players are clutch hitters?
- Ex 2: Predicting winning records for the 2026 season

Final projects!

You will give a 2 minute (one slide) presentation on your project during class on April 20th

Final project written reports are due at 11pm on April 26th

- Report should be a Jupyter notebook approximately 7 pages long
- A final project Jupyter notebook template will be uploaded to the class GitHub site

The baseball season starts this week!

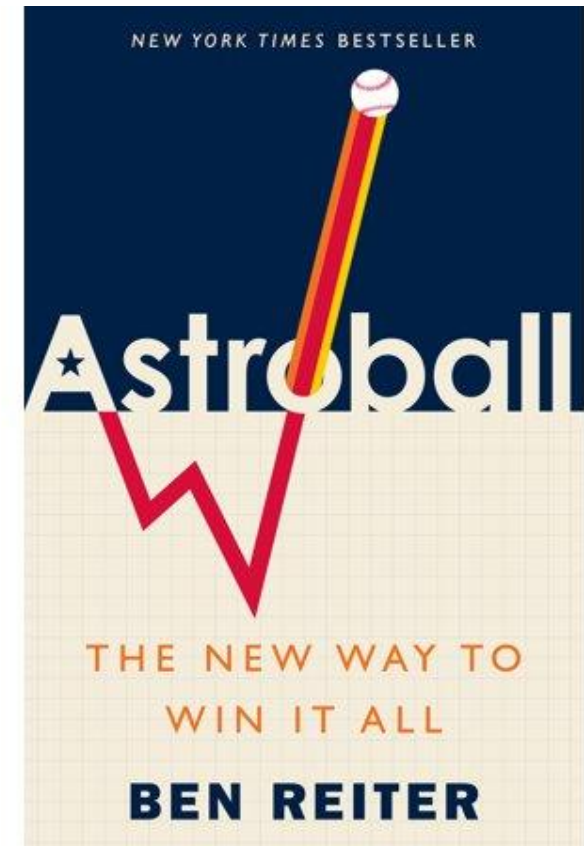


 Red Sox  Reds	Thu, Mar 26 4:10 PM	 Red Sox  Reds	Mar 28 4:10 PM
 Red Sox  Reds	Mar 29 1:40 PM	 Red Sox  Astros	Mar 30 8:10 PM
 Red Sox  Astros	Mar 31 8:10 PM	 Red Sox  Astros	Apr 1 2:10 PM
 Padres  Red Sox	Apr 3 2:10 PM	 Padres  Red Sox	Apr 4 4:10 PM

Discussion on chapter 6 of Astrobball

Let's discuss the chapter 6 for 3 minutes in small groups of 3-4 people and then have a larger conversation as a group

- E.g., Discuss your quote and reaction to chapter 6



Sports Illustrated cover jinx hitting 2014 Astros?



MLB

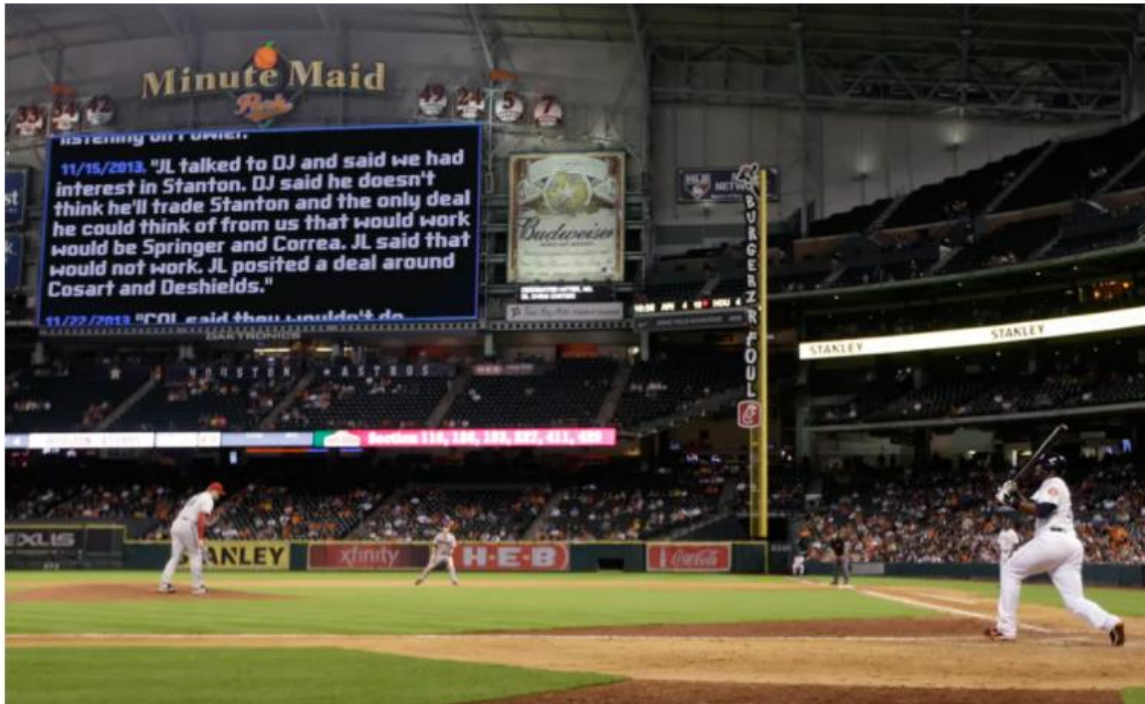
Leaked: 10 Months Of The Houston Astros' Internal Trade Talks



Barry Petchesky
6/30/14 1:19PM



17



Two years ago, the Houston Astros constructed "Ground Control"—a built-from-scratch online database for the private use of the Astros front office.

It is by all accounts a marvel, an easy-to-use interface giving executives instant access to player statistics, video, and communications with other front offices around baseball.

All it needs, apparently, is a little better password protection.

What Happened to the Houston Astros' Hacker?



As the 2017 spring season wore on, it became increasingly difficult to score against one of the four clubs in the softball league at the federal prison camp in Cumberland, Md.

The Dogs played the standard 10 fielders, but it seemed as if they had twice as many....

[The bail project](#)



J.D. Martinez

Positions: Outfielder and Designated Hitter

Bats: Right • Throws: Right

6-3, 230lb (190cm, 104kg)

Born: [August 21, 1987](#) (Age: 38-214d) in Miami, FL 

[More bio, uniform, draft, salary info ▼](#)

SUMMARY	WAR	AB	H	HR	BA	R	RBI	SB	OBP	SLG	OPS	OPS+
Career	30.8	6152	1741	331	.283	897	1071	26	.348	.516	.863	131

Martinez acknowledged that every athlete who has ever been cut from a team feels aggrieved.

What hurt most was how the Astros had done it.

"They had all this data, all these nerds and geeks, and I think they forgot that at the end of the day, everyone is still human," he said.

"And a human can adapt, and a human can adjust."

Probability models

Probability

Probability is a way of measuring the likelihood that an event will occur

Definitions:

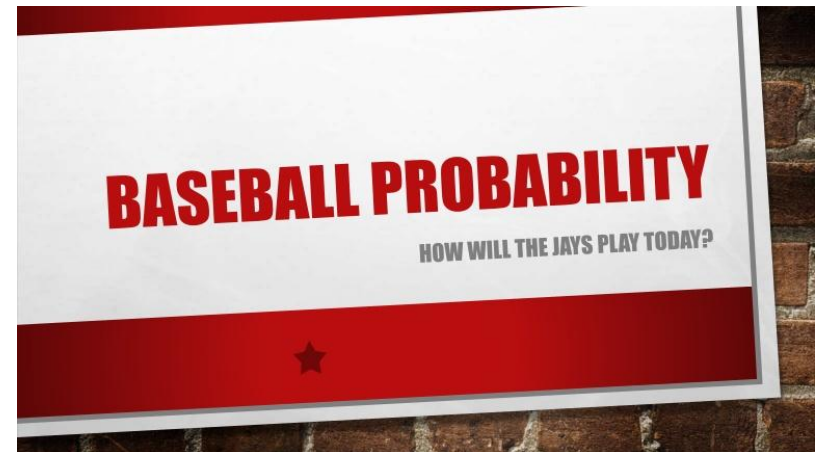
Sample space: All possible outcomes that can occur

- E.g., all outcomes that could happen as an outcome of a plate appearance

Event: A subset of the sample space

- E.g., A hit is {single, double, triple, home run}

Probability models assigns a number between 0 and 1 to the outcome of an event occurring



Random variables

Random variables map random events to numbers

- E.g., we could call a hit = 1, and an out = 0
 - Sample space $\{0, 1\}$
- Random variables can be continuous or discrete numbers

Typically, random variables are denoted with capital letters can take on values in the sample space

- E.g., X , or Z

We write the probability that a random variable X will take on a value between two other numbers: **$P(a < X < b)$**



Same properties of probability apply:

- $P(a < X < b) \geq 0$
- $\sum P(X = x_i) = 1$

Parametric probability models

We have explored probability using:

- Probability rules to calculate the probability of an event
- Dice/spinners (and Python) to simulate random events

When using random variables, we can use mathematical functions to calculate the probability of random variables taking on particular values

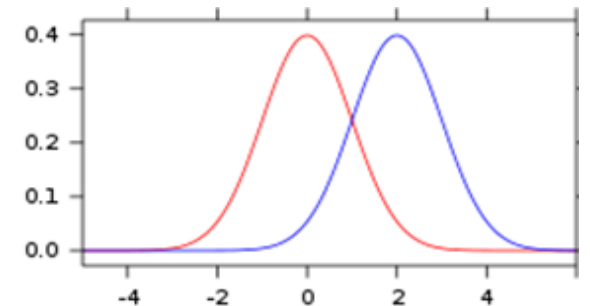
$$\bullet P(\mathbf{X}_1 = x_1, \mathbf{X}_2 = x_2, \mathbf{X}_3 = x_3) = f(x_1, x_2, x_3)$$

Often the functions have a set of parameters that we need to fix to get a specific probability model

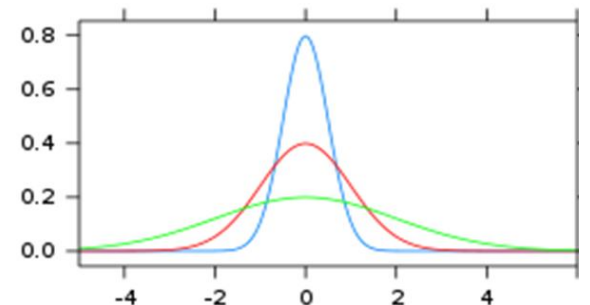
$$\bullet P(\mathbf{X} = x) = f(x; \theta)$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Changing μ



Changing σ



Probability model terminology

Probability mass functions (PMFs) are functions that help us calculate probabilities for discrete random variables

- E.g., random variables that take on integer values: 1, 2, 3, ...

Probability density functions (PDFs) are functions that help us calculate probabilities for continuous random variables

- E.g., random variables that take on real values

Cumulative distribution functions (CDFs) give us the probability that a random variable X will be within a particular range of values

Probability mass function (PMF)

$$P(X \leq x) :$$

$$= \sum_{x_i < x} f(x_i)$$

Probability density function (PDF)

$$P(a < X \leq b) = \int_a^b f(x) dx$$

Bernoulli Distribution

The **Bernoulli Distribution** models the probability of two outcomes:

- Success: $X = 1$
- Failure: $X = 0$
- E.g., getting a head ($X = 1$) or a tail ($X = 0$) for flipping a coin

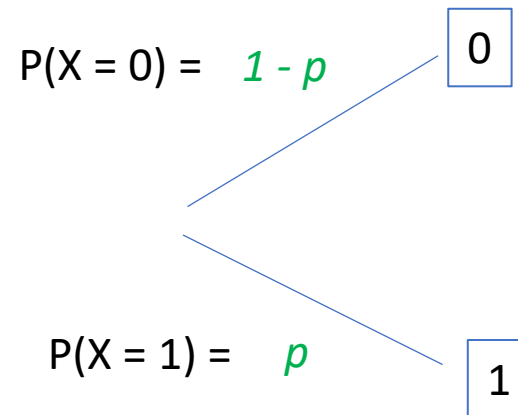
Model has one **parameter p** , which is the probability of getting a 1

- E.g., the probability of getting head on a coin flip
- $P(X = 1) = ?$ $P(X = 0) = ?$

Bernoulli Distribution

We can you draw the Bernoulli distribution as a tree diagram...

- $P(X = x; p)$



Bernoulli Distribution

As we discussed, **the probability mass functions** is a function that gives the probability of each outcome in the sample space of a *discrete distribution*

For Bernoulli Distributions we can write the probability mass function as:

$$f(k; p) = \begin{cases} p & \text{if } k = 1 \\ 1 - p & \text{if } k = 0 \end{cases} \quad f(k; p) = p^k(1 - p)^{1-k} \quad \text{for } k \in \{0, 1\}$$

We can also plot the probability mass function

- If $p = .5$ what would it look like?
- If $p = .9$ what would it look like?

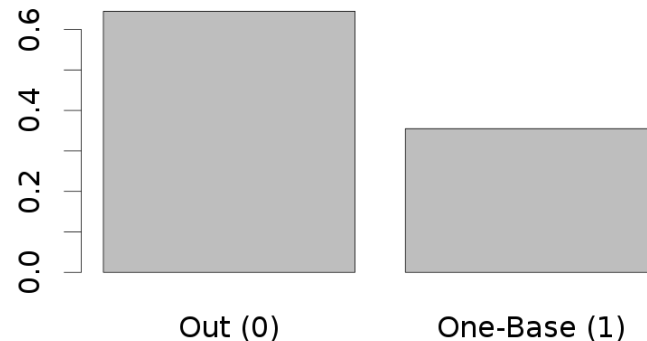


Parameter p

Bernoulli Distribution

Can you think of baseball example that can be modeled using a Bernoulli distribution?

- Example: The probability that a player gets on base for a given at bat
- If we were modeling Shohei Ohtani, what would a good estimate of p be?
 - OBP = .374, so a good number to use would be $p = .374$



Binomial distribution

The **Binomial Distribution** models the probability of having k successes out of the n trials

- Where the probability of success on each trial is p

Example: if a player comes to bat $n = 4$ times in a game, what is the probability of getting on-base k times

- What is the sample space?

Assumes the same probability of getting on-base each plate appearance (p)

- Are the assumptions reasonable for this model?
 - No streakiness

Tree diagrams of binomial distribution

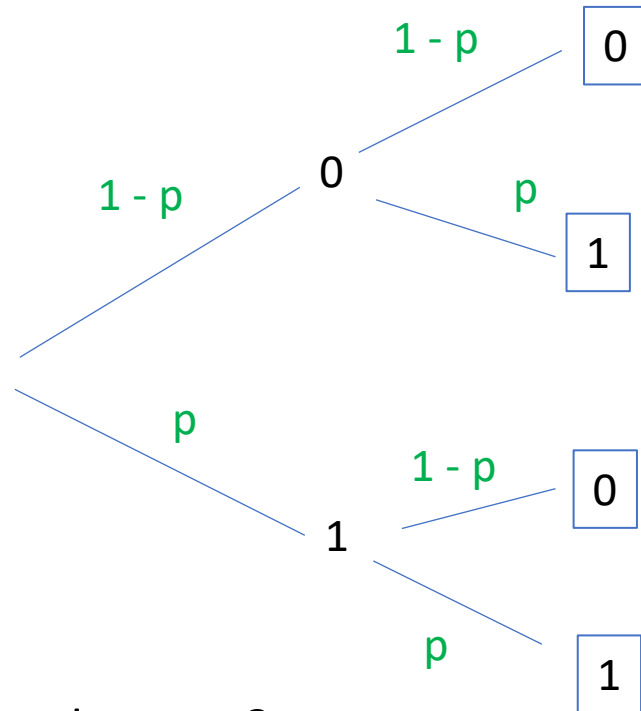
The binomial distribution can be thought of sequences of Bernoulli trials:

Binomial distribution for $n = 2$:

$$P(X = 0) = (1-p)^2$$

$$P(X = 1) = 2 \cdot p \cdot (1-p)$$

$$P(X = 2) = p^2$$



If $n = 2$, what is the sample space?

A: $\{0, 1, 2\}$

$$P_{\text{Bernoulli}}(X_1 = 0; p) \cdot P_{\text{Bernoulli}}(X_2 = 0; p) = (1-p)^2$$
$$= P_{\text{Binomial}}(X = 0; n = 2, p) = (1-p)^2$$

$$P_{\text{Bernoulli}}(X_1 = 0; p) \cdot P_{\text{Bernoulli}}(X_2 = 1; p) = p \cdot (1-p)$$

$$P_{\text{Binomial}}(X = 1; n = 2, p) = 2 \cdot p \cdot (1-p)$$

$$P_{\text{Bernoulli}}(X_1 = 1; p) \cdot P_{\text{Bernoulli}}(X_2 = 0; p) = p \cdot (1-p)$$

$$P_{\text{Bernoulli}}(X_1 = 1; p) \cdot P_{\text{Bernoulli}}(X_2 = 1; p) = p^2$$

$$= P_{\text{Binomial}}(X = 2; n = 2, p) = p^2$$

Binomial distribution

Probability of getting k successes out of n trials is:

Parameters: p and n


$$P(X = k) = f(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

What values can X take on (i.e., what is the sample space)?

- k ranges from 0, 1, ..., n

n choose k

N choose k function tells us how many ways there are to select ***k*** items out of ***n*** total

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Q: How many ways are there to choose 3 things out of a total of 8?

A: $(8 \cdot 7 \cdot 6) / (3 \cdot 2) = 56$

Binomial distribution

In 2013, Koji Uehara struck out about 33% of the batters he faced

If he pitched to 5 batters in a game, use the binomial distribution to calculate the probability he would strike out 2 of them?

$$P(X = k) = f(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Binomial distribution

Koji Uehara typically strikes out about 33% of the batters he faces

If he pitched to 5 batters in a game, use the binomial distribution to calculate the probability he would strike out 2 of them?

What are the parameters?

$$n = 5, \quad p = .333 \quad k = 2$$

$$P(X = k) = f(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$P(X = 2) = f(k; 5, .333) = \binom{5}{2} .333^2 (1 - .333)^3$$

$$= .329$$

Binomial distribution

Suppose a Uehara pitched to 20 batters

How could we calculate the probability that he would get more than 10 strike outs?

$$P(X > 10) = \sum_{k=11}^{20} f(k; n = 20, p = .333)$$

$$= \sum_{k=11}^{20} \binom{20}{k} .333^k (1 - .333)^{20-k}$$

$$= .0376$$

n choose k in Python

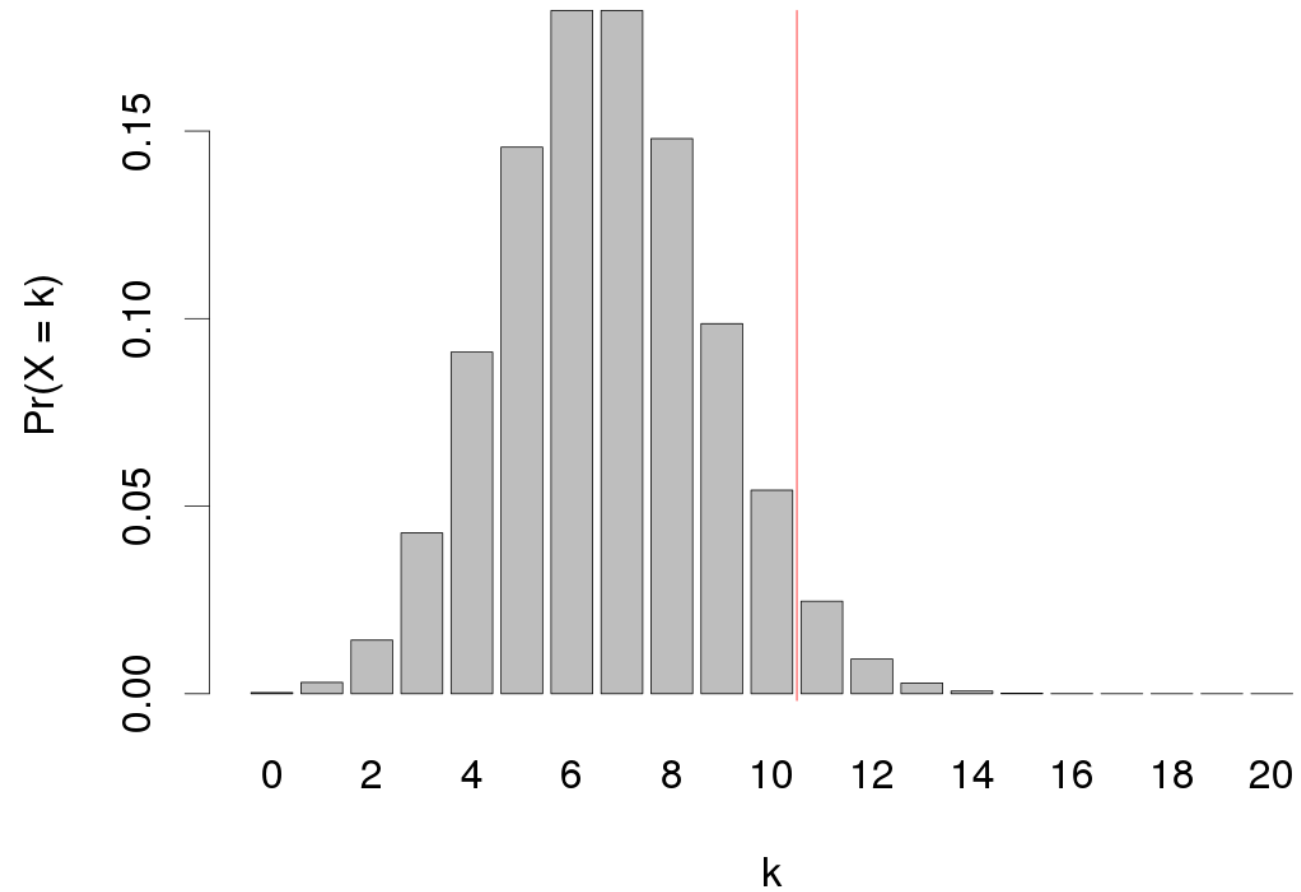
import math

n = 20

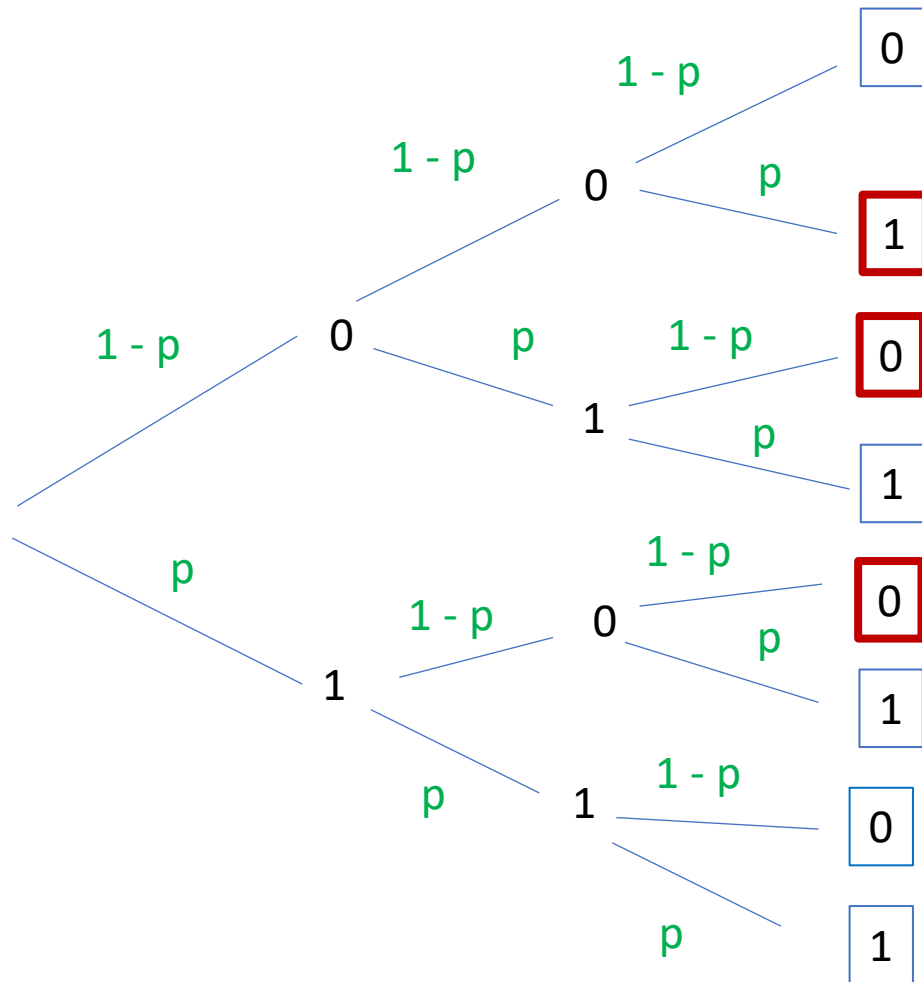
k = 10

math.comb(n, k)

Binomial distribution



Verify binomial distribution agrees with tree diagrams



Checking for:

$$P(X = 1; n = 3, p)$$

What are the possible sequences for $X = 1$?

- 0, 0, 1
- 0, 1, 0,
- 1, 0, 0

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Binomial distribution in Python

create the binomial pmf

```
from scipy.stats import binom
```

```
binom.pmf(k_sequence, n, p)
```

Can use a sequence such as range(n)

returns an ndarray of values

generate random values from a binomial distribution

```
import numpy as np
```

```
np.random.binomial(n, p, size)
```

Generates random
numbers in range 0 to n

How many random numbers to
generate

Lab 7: Comparing binomial distribution to real data

On lab 7 you will....

1. Calculate the frequency that Aaron Judge get 0, 1, 2, 3 or 4 hits in games where he had 4 at-bats
2. Compare this to the predicted frequencies of getting 1, 2, 3 or 4 hits in games where he had 4 at-bats
 - This will test if the assumptions that each hit is independent and generated with the same rate (p) at each at-bat
3. Run a hypothesis test to assess whether there is a statistically significant difference between the binomial model and the observed frequencies

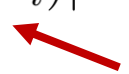
Lab 7: Comparing binomial distribution to real data

To run a hypothesis test comparing binomial probabilities to observed frequencies you will be testing the hypotheses:

- H_0 : $p_{0\text{-observed}} = p_{0\text{-binomial}}, p_{1\text{-observed}} = p_{1\text{-binomial}}, \dots, p_{4\text{-observed}} = p_{4\text{-binomial}},$
- H_A : At least one of the observed proportions is different than the binomial value

You will use the Total Variation Distance (TVD) statistic:

$$TVD(p, q) = \frac{1}{2} \sum_i |p(x_i) - q(x_i)|$$

Observed frequencies  Values from the binomial PMF 

You will calculate the TVD on the real data vs. binomial distribution

You will then generate random data from a binomial distribution to see if the real TVD looks similar to the randomly generated TVDs

Brief mention: Expected values

The **expected value** of a random variable (denoted $E[X]$) is a weighted average of all possible outcomes

The expected value is calculated using:

$$E[X] = \sum_i x_i f(x_i)$$


Discrete RV: Probability mass function (PMF)

$$E[X] = \int x f(x) dx$$


Continuous RV: Probability density function (PDF)

We can also calculate conditional expected values, which is the expected value given another particular outcome has occurred

$$E[X|Y = y] = \sum_i x_i P(X = x_i|Y = y)$$

Brief mention: Expected values

Expected value of a Bernoulli random variable:

$$0 \cdot (1-p) + 1 \cdot (p) = p$$

$$E[X] = \sum_i x_i f(x_i)$$

Expected value of a Binomial random variable:

$$0 \cdot 1 \cdot (1-p)^n + 1 \cdot n \cdot (1-p)^{n-1} \cdot p + \dots + n \cdot 1 \cdot (p)^k$$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E[X] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}$$

Run expectancy matrix

A **Run Expectancy Matrix** is a table that gives the average number of runs scored given base-out state and the end of an inning

- $E[\text{Runs} \mid \text{runners on base, \# outs}]$
- E.g., $E[\text{Runs} \mid \text{runner on first and second with 2 outs}]$

The runner positions can be denoted with a three digit binary number

- $E[\text{Runs} \mid 011-2]$



Runner on 1st and 2nd base with 2 outs

	0 Outs	1 Out	2 Outs
000			
001			
010			
011			
100			
101			
110			
111			

You will create this in lab 7 part 3!

Guest speaker Sam Larson

Sam Larson is a Senior Analyst in Biomechanics and Baseball Science for the Boston Red Sox

He will tell us about his experience working for an MLB team

